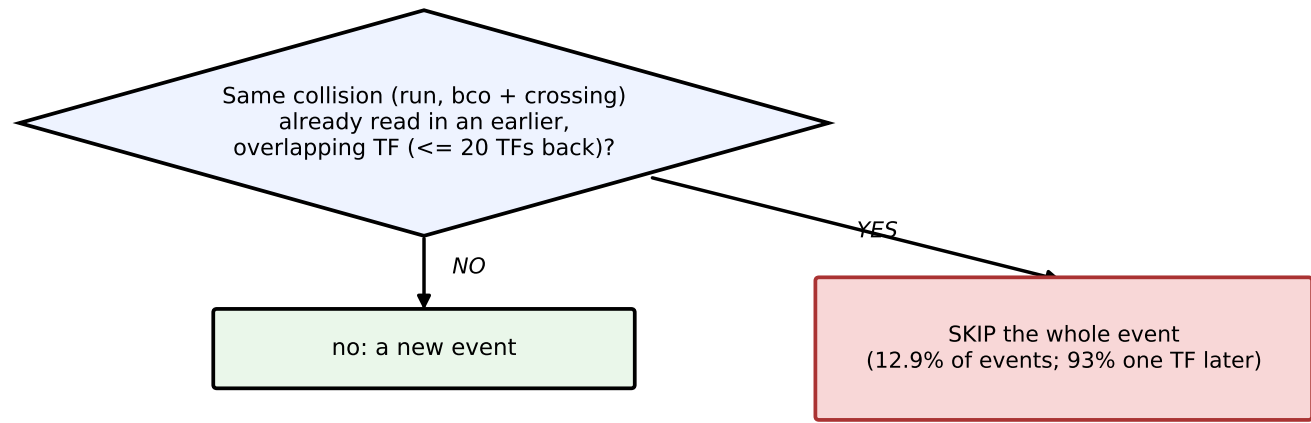


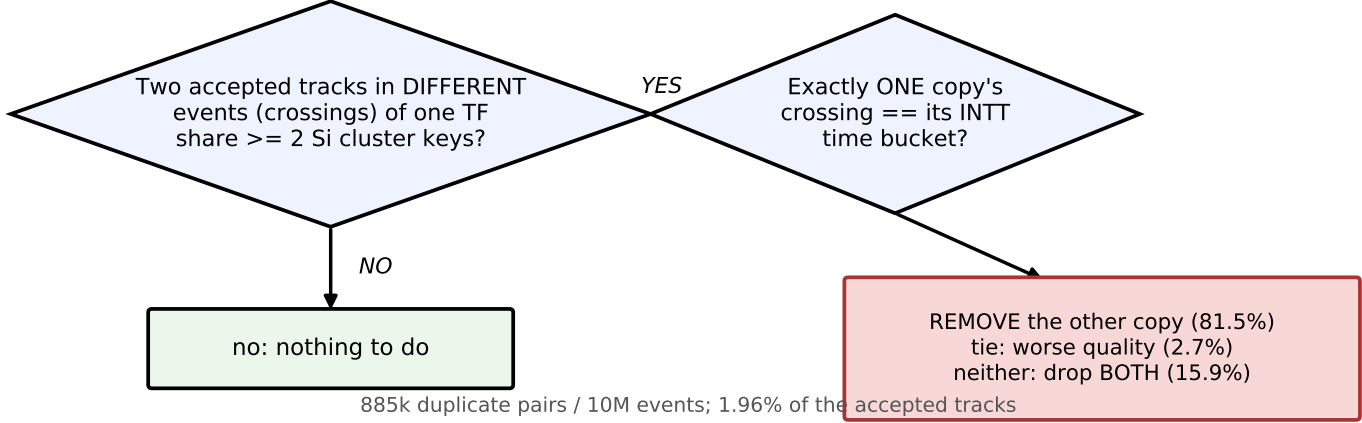
Split-track and duplicate treatment, ana573 default, page 1: events and tracks -- every decision in processing order

STAGE 0 -- pre-passes over all events, before the event loop (event = one crossing with a vertex; the DSTs hold many events per trigger frame, TF)

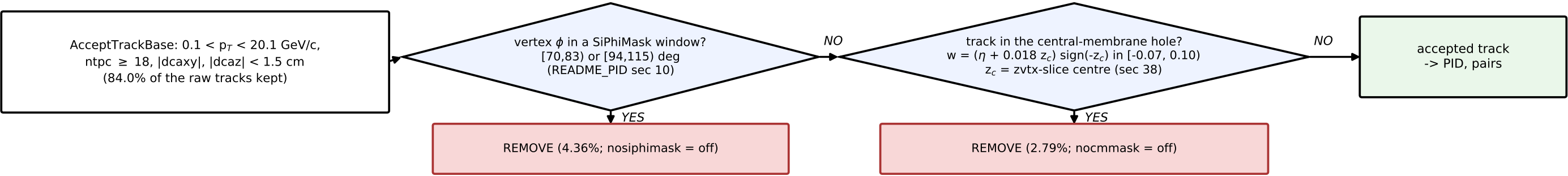
(a) BuildTFDupRows (noTFdup = off)



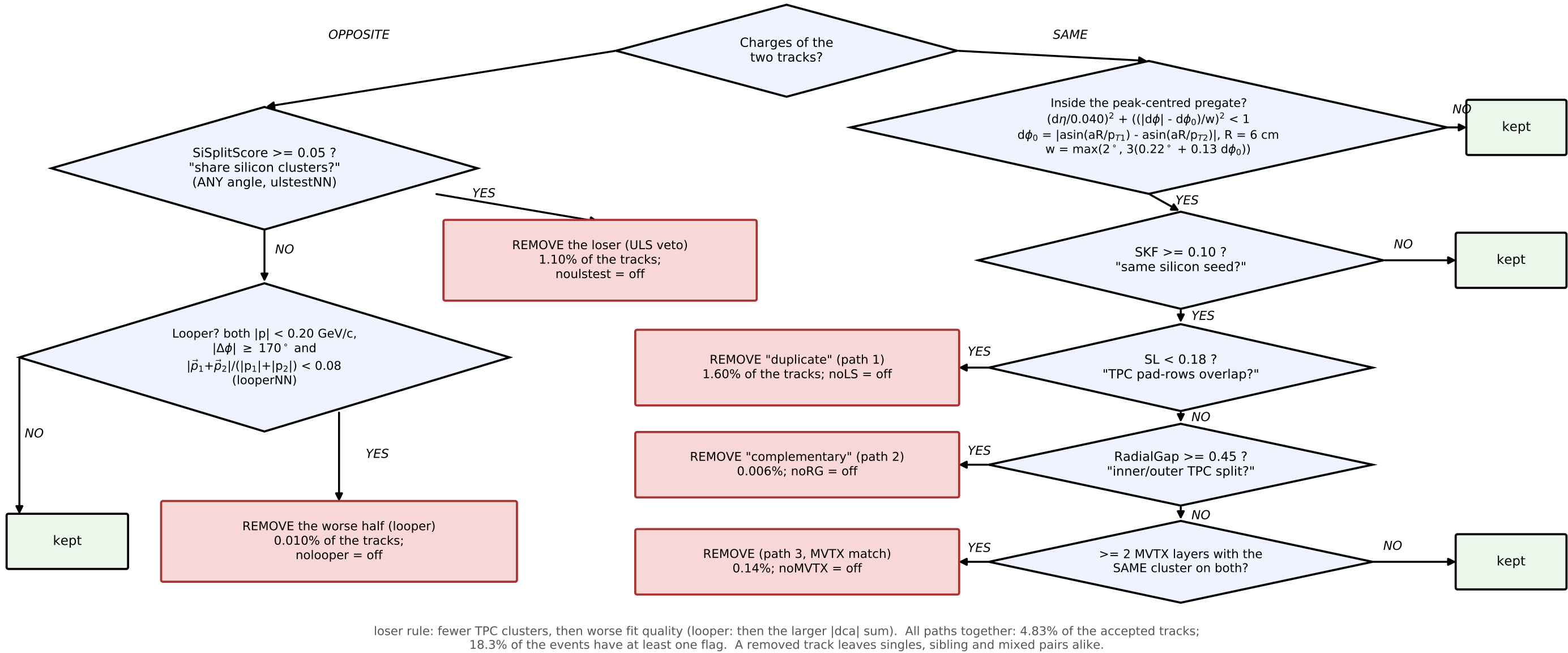
(b) BuildXTFLosers (noXTF = off, XTFclean = non-strict)



STAGE 1 -- per track (AcceptTrack)

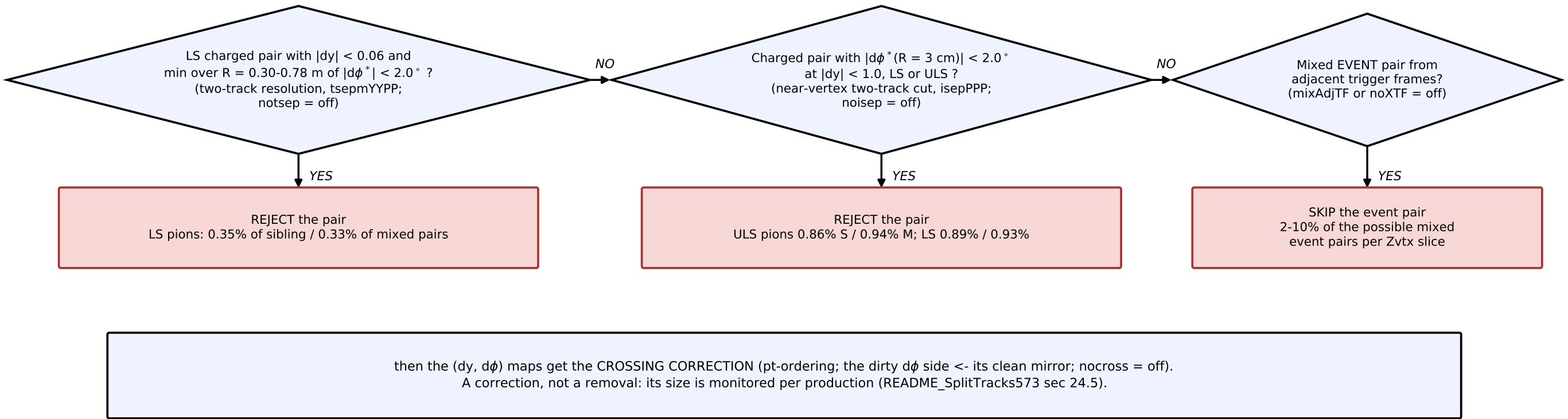


STAGE 2 -- per event (one crossing), every pair of accepted charged tracks (corral\_loop.cxx split-track pre-pass)



Opposite-sign candidates are every accepted charged track (not only pions); daughters of V0s in their mass peak stay out (README\_PID sec 11).  
Pair-level cuts, mixing, the crossing correction and the watch items: page 2. Pictures of each case: ana573/SplitTrackTypes\_ana573.pdf.

STAGE 3 -- pair level (CalcRm::PairInfo): sibling AND mixed pairs alike, in this order; then the maps



WATCH -- monitored in every production, not removed (user's rule: monitor, do not fix)

The  $\pm 15^\circ$  ULS notch (secs 32-34)  
a  $\sim 4\%$  deficit of ULS SIBLING pion pairs at  $10\text{-}20^\circ$  vertex  $d\phi$  ( $\sim 12\%$  in R2 at  $|dy| < 0.1$ ). Cause: pairs whose tracks MEET in  $\phi$  at  $R_m \approx 0.10\text{-}0.30$  m (INTT-TPC gap) are lost in the silicon-TPC matching; no Corral cleaner makes it. For tracking.  
Page: "The  $\pm 15^\circ$  ULS notch: WATCH" (S/M vs  $d\phi$ ,  $R_m$ ,  $p_T$ ,  $N_{ch}$ ).

Also monitored

- the crossing losses of each production (ana573 50M, sec 24.5:  $+5.6\text{e-}3$  pairs/event  $\pi^+\pi^-$ ,  $3.7\text{-}10\times$  ana532)
- the same-TF share of mixed pairs (5.8%, sec 18.20)
- V0 masses and C(Q) peak positions (momentum scale: a few % in the field fixes them)
- tracks removed per cleaner: the split-removal page and the log

For tracking, beyond the notch: the  $\pi^+/\pi^-$  split asymmetry (secs 15.2, 19); the SiPhiMask zones, the  $90^\circ$  pattern of  $\langle |dcaxy| \rangle$  vs  $\phi$  and the dcaz ramp in  $\eta$  (README\_PID sec 10).